

SAANVI REDDY BARADI

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SUMMARY

AI Engineer with production-grade experience building end-to-end generative AI systems, from design through deployment. Proficient in LangChain, Vertex AI (Gemini), RAG architecture, LLM evaluation, and agentic workflows. Skilled in designing scalable APIs, implementing safety validation frameworks, monitoring production systems, and optimizing performance at scale. Seeking AI Engineer roles to build reliable, production-ready AI-powered products with strong software engineering practices.

EDUCATION

University of South Florida, Tampa, FL

Master of Science, Artificial Intelligence and Business Analytics

Aug 2024 – May 2026

GPA: 3.95 / 4.0

Coursework: Machine Learning, Business Statistics, Advanced Database Management Systems, Data Visualization, Big Data Analytics, Data Warehousing, Application Development for Analytics

Malla Reddy Engineering College for Women, Hyderabad, India

Bachelor of Technology, Computer Science Engineering - Data Science

Jun 2020 – Apr 2024

GPA: 3.7 / 4.0

TECHNICAL SKILLS

Generative AI & LLMs: LangChain (Chains, Agents, LCEL), Vertex AI (Gemini), OpenAI API, Prompt Engineering, RAG Pipelines, Vector Search, pgvector, Embeddings, LLM Evaluation (BLEU, ROUGE), Hallucination Detection & Mitigation.

AI/ML Engineering: Scikit-learn, PyTorch, XGBoost, SHAP, SMOTE, Ensemble Methods (Random Forest, Gradient Boosting, SVM), NLP (NLTK, spaCy, TF-IDF, NER), Feature Engineering, Cross-Validation, Hyperparameter Tuning

Software Engineering & APIs: FastAPI, RESTful API Design, Pydantic, Error Handling, Rate Limiting, pytest, Python, SQL, R, Git/GitHub

Production & MLOps: Docker & Docker Compose, Model Serving, Streamlit, Cloud Logging, Data Validation Pipelines, Performance Optimization, Containerization

Cloud Platforms: Google Cloud Platform (Vertex AI, Cloud Run, Cloud Logging, BigQuery)

BI & Visualization: Tableau, Power BI (DAX), Matplotlib, Seaborn

EXPERIENCE

Project Assistant - University of South Florida

Feb 2025 – Sep 2025

- Designed and deployed a Python-based ETL pipeline standardizing 180+ country name variations (EU, USSR, UAE) into ISO codes using fuzzy matching (RapidFuzz) and custom override dictionaries, reducing data mismatch errors by ~90%
- Automated end-to-end data processing workflows ingestion, matching, standardization, validation, and mismatch flagging cutting manual data preparation time by an estimated 70%
- Integrated U.S. Census Bureau ACS 5-Year API using FIPS-based conditional logic to extract demographic and income data at ZIP, county, and state granularity, producing warehouse-ready outputs consumed by Tableau BI dashboards
- Designed data validation frameworks and automated quality checks feeding downstream ML feature pipelines consumed by 3+ analytics teams

Data Science Intern - Accenture

Jun 2023 – May 2024

- Built and optimized ML data pipelines for cognitive and socioeconomic datasets (5M+ records), feature engineering, EDA, and data quality resolution directly improving model performance across 5+ analytics teams
- Developed Tableau and Power BI dashboards translating ML model outputs into actionable KPIs for C-level stakeholders; automated reporting reducing manual reporting time by 30%
- Built SQL-based reporting pipelines analyzing operational trends and improving feature consistency across ML workflows; designed and documented data models for 8+ downstream analytics use cases

Research Assistant – Malla Reddy Engineering College for women

Jun 2022 – Jun 2023

- Analyzed 18,617 real OSHA construction accident reports using NLP techniques - cleaned and processed raw text, then trained 4 machine learning models to predict accident risk level; best model (SVM) achieved 96% accuracy and AUC-ROC of 0.9867
- Used spaCy to automatically pull-out key details from accident reports - equipment involved, hazard type, and location; connected Gemini LLM to suggest immediate safety actions and relevant OSHA standards for high-risk cases
- Built and deployed an interactive dashboard using Streamlit and Docker where users can paste any accident report and instantly see the risk level, key risk factors, and safety recommendations
- Findings from the model helped identify the most dangerous patterns across construction sites, contributing to safety improvements that reduced reported workplace incidents by 20%

PROJECTS

AI-Powered SQL Analytics Assistant | LangChain, Vertex AI (Gemini), RAG, Python, SQL

- Built production-ready NL-to-SQL engine with schema-aware prompting, leveraging live database metadata to convert natural language questions into context-aware, validated SQL queries
- Engineered automated SQL validation guardrails (SELECT-only enforcement, safe join patterns, LIMIT compliance) with modular validation layer preventing unsafe query execution
- Developed Streamlit based self-service analytics UI with interactive query results, automated summaries, and visualizations for non-technical end-users
- Architected modular LLM pipeline with Vertex AI Gemini integration, local fallback support, and cost-aware AI design reflecting real-world production constraints

RAG-based Document Q&A API Service | LangChain, FastAPI, PostgreSQL (pgvector), Vertex AI, Pytest

- Built production-ready RAG service: document ingestion (PDF/text parsing), embedding generation (Vertex AI), vector storage (PostgreSQL pgvector), retrieval-augmented generation
- Designed FastAPI endpoints with Pydantic validation, error handling, rate limiting, and Cloud Logging for production monitoring; achieved <100ms response latency
- Implemented LLM evaluation framework measuring answer relevance (ROUGE), fact consistency, retrieval quality; pytest unit/integration tests with performance benchmarks

CERTIFICATIONS

Data Science Fundamentals with Python and SQL – IBM | Introduction to Data Science – IBM | Microsoft Power BI Data Analyst (PL-300) | Programming Essentials in Python – Cisco (OpenEDG) | Python (Basic) – HackerRank